A detailed description of the first three steps of the proposed approach

Below, we describe the first steps of our seven-step data-driven methodology. Preprocessed resting-state functional magnetic resonance imaging (rs-fMRI) data were analyzed using NeuroMark-guided spatially constrained independent component analysis (scICA) to estimate subject-specific intrinsic connectivity networks (ICNs) and their corresponding time courses. Unlike conventional group ICA based on temporal concatenation, the NeuroMark-guided MOO-ICAR approach estimates individualized spatial maps while using reference templates to maintain correspondence across subjects [5, 2]. Figure 1 illustrates in detail how scICA (also referred to as CICA in the literature) operates.

Step 1: NeuroMark-guided spatially constrained ICA

Let n denote the number of subjects, T_i the number of time points, and V the number of voxels for subject i, where $i = 1, \ldots, n$. The preprocessed rs-fMRI data for subject i are represented by the matrix

$$Y_i \in R^{T_i \times V}. \tag{1}$$

Let K denote the number of NeuroMark reference components retained for analysis. The fixed NeuroMark template is represented by

$$S_{\mathrm{NM}} = \begin{bmatrix} s_1^{\mathrm{NM}} \\ \vdots \\ s_K^{\mathrm{NM}} \end{bmatrix}_{K \times V}, \tag{2}$$

where $s_k^{\mathrm{NM}} \in R^{1 \times V}$ denotes the reference spatial map for component k. The template provides spatial prior information to guide component estimation; it is not assumed to be identical to the spatial maps estimated for individual subjects. For each subject, the MOO-ICAR algorithm estimates individualized spatial maps by optimizing component independence and correspondence to the respective NeuroMark reference maps [2]. Let $X_i \in R^{Q_i \times V}$ denote the whitened and dimension-reduced data representation used internally by the ICA algorithm, where Q_i is its retained dimension. The k-th subject-specific component is represented as

$$\widehat{s}_{ik} = \widehat{w}_{ik}^{\mathsf{T}} X_i, \tag{3}$$

where $\widehat{w}_{ik} \in R^{Q_i}$ is the estimated unmixing vector and $\widehat{s}_{ik} \in R^{1 \times V}$ is the resulting subject-specific spatial map. The independence objective is based on a non-Gaussianity criterion, which can be expressed schematically as

$$J(\widehat{s}_{ik}) = \left\{ E\left[G(\widehat{s}_{ik})\right] - E\left[G(v)\right] \right\}^2, \tag{4}$$

where G is a nonquadratic function, v is a standard Gaussian random variable, and $E[\cdot]$ denotes expectation over the spatial samples. This objective encourages the estimated component to depart from Gaussianity, supporting the identification of statistically independent sources. The spatial correspondence objective is expressed as

$$F(\widehat{s}_{ik}, s_k^{\mathrm{NM}}) = E\left[\widehat{s}_{ik} s_k^{\mathrm{NM}}\right], \tag{5}$$

where the template and estimated map are compared using the normalization adopted by the algorithm. This objective encourages the estimated component to correspond to its reference spatial pattern while preserving subject-specific variation. The MOO-ICAR estimation can therefore be summarized as the joint optimization of independence and spatial correspondence:

$$\begin{aligned} \max_{\widehat{w}_{ik}} \quad & \left(J(\widehat{s}_{ik}), F(\widehat{s}_{ik}, s_k^{\mathrm{NM}})\right) \\ \text{subject to} \quad & \|\widehat{w}_{ik}\|_2 = 1, \end{aligned} \tag{6}$$

with the objective combination and optimization settings determined by the MOO-ICAR implementation. Equations (4)–(6) provide a schematic mathematical description of the template-guided estimation rather than a replacement for the implementation-specific optimization routine. The resulting individualized spatial maps are assembled into

$$\widehat{S}_i = \begin{bmatrix} \widehat{s}_{i1} \\ \vdots \\ \widehat{s}_{iK} \end{bmatrix}_{K \times V}. \tag{7}$$

In general,

$$\widehat{S}_i \neq S_{\mathrm{NM}}, \tag{8}$$

because the NeuroMark template guides the estimation rather than imposing fixed spatial maps on every subject. This distinction allows individualized functional-network patterns to be estimated while maintaining correspondence with the reference ICNs. The estimated spatial maps and corresponding component time courses can be represented through the linear mixture model

$$Y_i \approx \widehat{A}_i \widehat{S}_i + E_i, \tag{9}$$

where $\widehat{A}_i \in R^{T_i \times K}$ contains the estimated component time courses and $E_i \in R^{T_i \times V}$ represents residual variability. Equation (9) expresses the factorization of the observed data; it is not the template-guided ICA optimization objective itself. If component time courses are subsequently recalculated by least-squares projection onto the estimated spatial maps, the corresponding estimate is

$$\widehat{A}_i^{\mathrm{LS}} = \underset{A_i}{\operatorname{argmin}} \left\| Y_i - A_i \widehat{S}_i \right\|_F^2 = Y_i \widehat{S}_i^{\dagger}, \tag{10}$$

where $\widehat{S}_i^{\dagger}$ denotes the Moore—Penrose pseudoinverse. This expression is applicable when time courses are estimated by least squares conditional on the estimated maps; it should not be interpreted as the MOO-ICAR optimization procedure or reported as the implemented time-course estimator unless it matches the actual analysis. Standard ICA estimates statistically independent components from the data without an externally specified spatial reference. Its theoretical motivation is that mixtures of independent non-Gaussian sources tend to be more Gaussian than the original sources. ICA therefore seeks directions that recover statistically independent, non-Gaussian signals from their mixtures [3, 1]. Spatially constrained ICA extends this framework by incorporating prior spatial information, allowing estimated components to correspond to predefined functional networks while retaining individual differences [2, 4]. The NeuroMark template provides a consistent set of reference ICNs across seven functional domains: subcortical (SC), auditory (AU), sensorimotor (SM), visual (VI), cognitive control (CC), default mode (DM), and cerebellar (CB). Each estimated component retains its correspondence to the relevant reference network, facilitating subsequent network-level functional connectivity analyses.

Step 2: Dynamic functional network connectivity

Step 2 involves estimating time-varying functional connectivity between component time courses. For subject i, let $\widehat{A}_i^{(q)}$ denote the segment of the estimated time courses within sliding window q, comprising L consecutive time points. The dynamic functional network connectivity matrix is

$$\mathrm{dFNC}_i^{(q)} = \mathrm{corr}\left(\widehat{A}_i^{(q)}\right) \in R^{K \times K}, \tag{11}$$

where $\mathrm{corr}(\cdot)$ computes the pairwise correlations between the component time courses within the window. Repeating this procedure over the time series produces a sequence of windowed connectivity matrices for each subject. For example, if six components are retained, the dFNC matrix in each window has dimensions 6×6. The resulting sequence captures temporal variation in functional coupling between the estimated ICNs and constitutes the input to subsequent dynamic connectivity analyses, as illustrated in Figure 2.

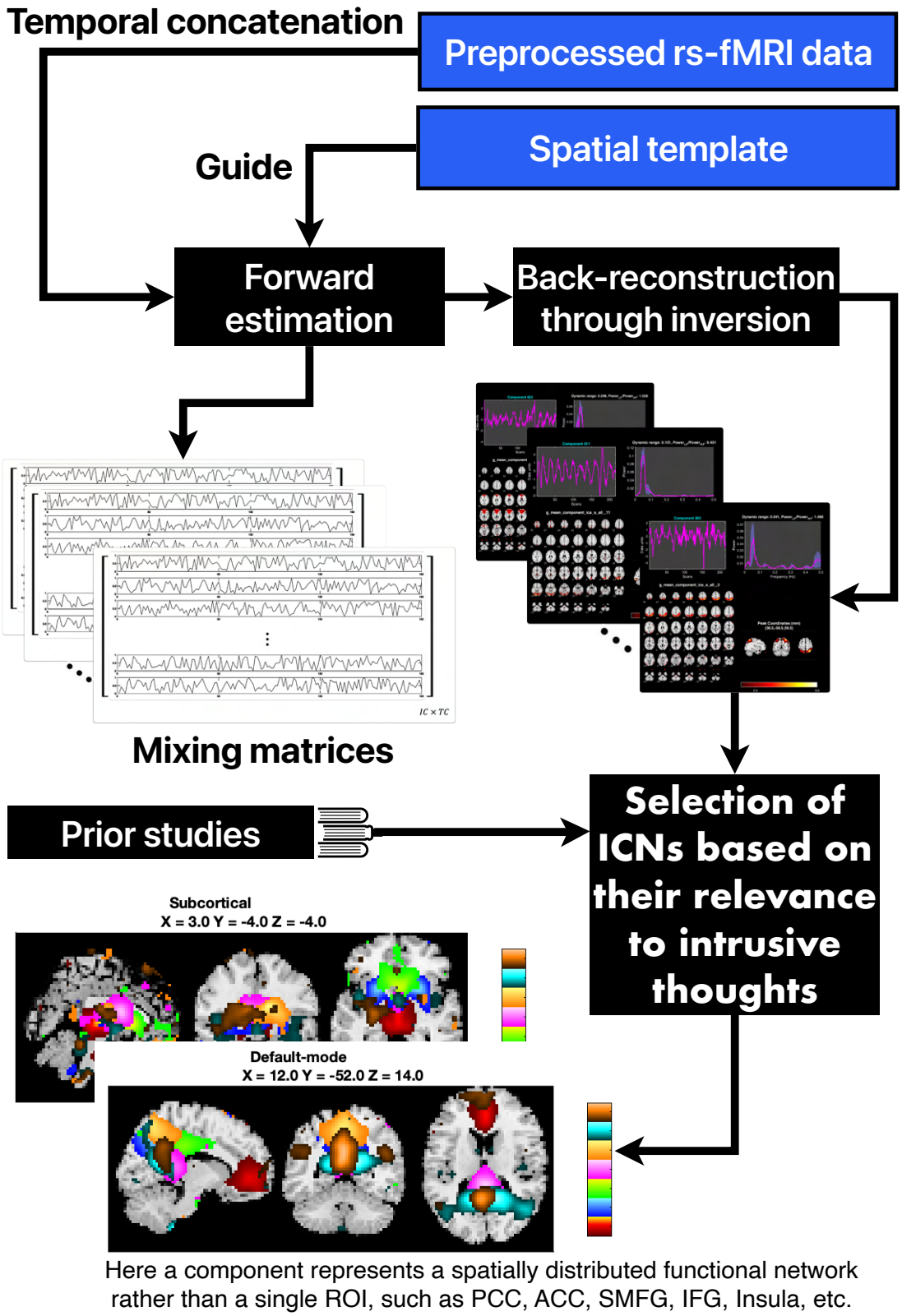


Figure 1: **NeuroMark-guided spatially constrained ICA (Step 1).** Preprocessed rs-fMRI data from each subject are decomposed using the MOO-ICAR framework, with NeuroMark reference maps guiding the estimation of individualized spatial components. The optimization balances component independence with spatial correspondence to the reference networks. The resulting subject-specific spatial maps and component time courses provide the inputs for subsequent dynamic functional network connectivity analysis. Subject-level estimation preserves individual variation without requiring temporal concatenation of participants or group-level ICA back-reconstruction. Healthy controls, when included, undergo the same subject-specific estimation procedure; their inclusion does not contribute to the estimation of another subject's maps.

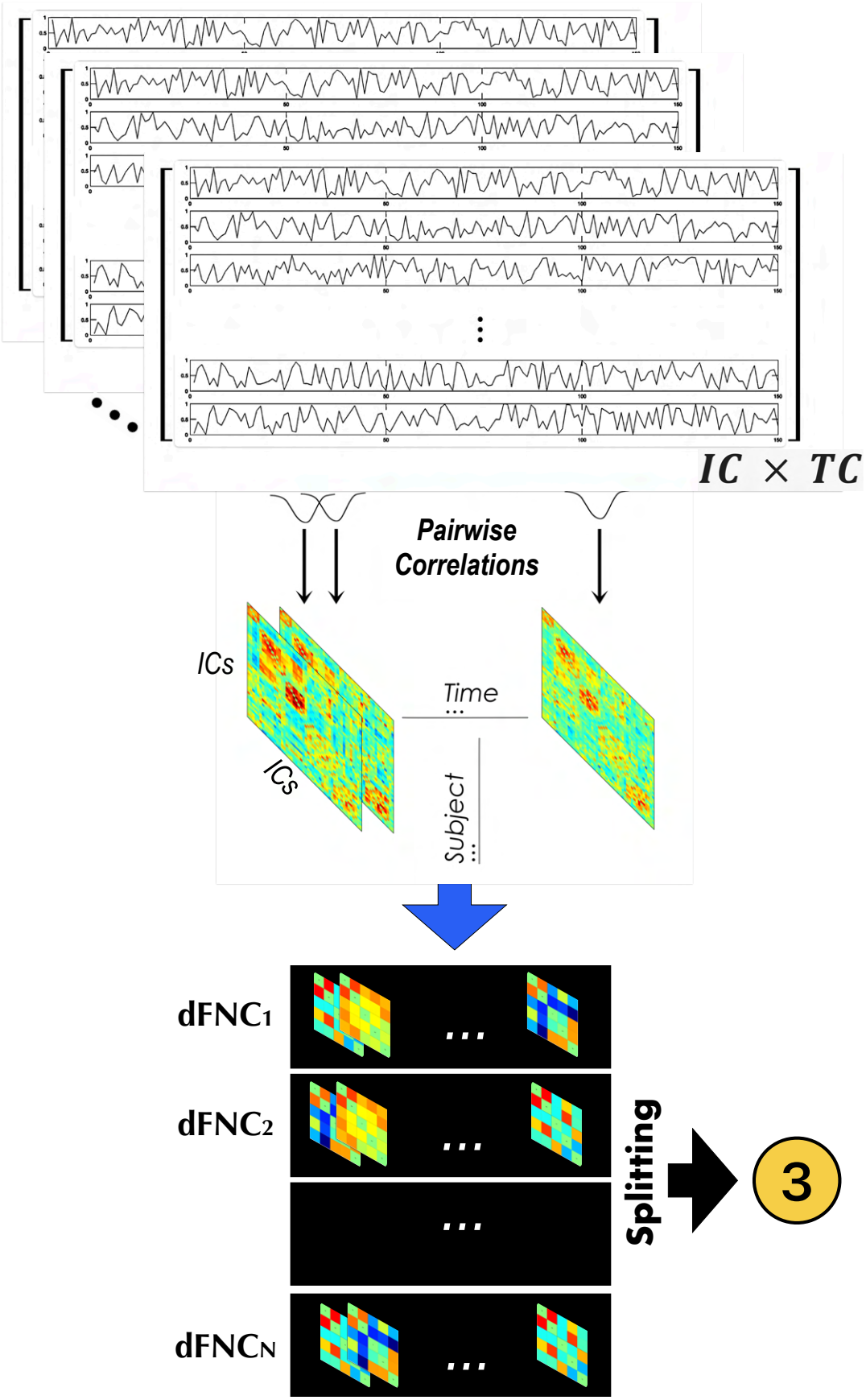


Figure 2: **Illustration of dFNC sequences across N subjects.** Matrices containing time series symbols are obtained from forward estimation of CICA (Figure 1). The time courses of the components of the selected ICNs—obtained from Step 1—are subjected to pairwise correlation of variable 1 and variable 2 across components, e.g., $IC-1$ and $IC-2$ for subject 1 correlated across time within window 1, and so on, using a sliding window algorithm to extract dynamic FNC matrices (dFNCs). The resulting dFNC sequences ($dFNC_1$ to $dFNC_n$) are then used for clustering, as illustrated in Figure 3.

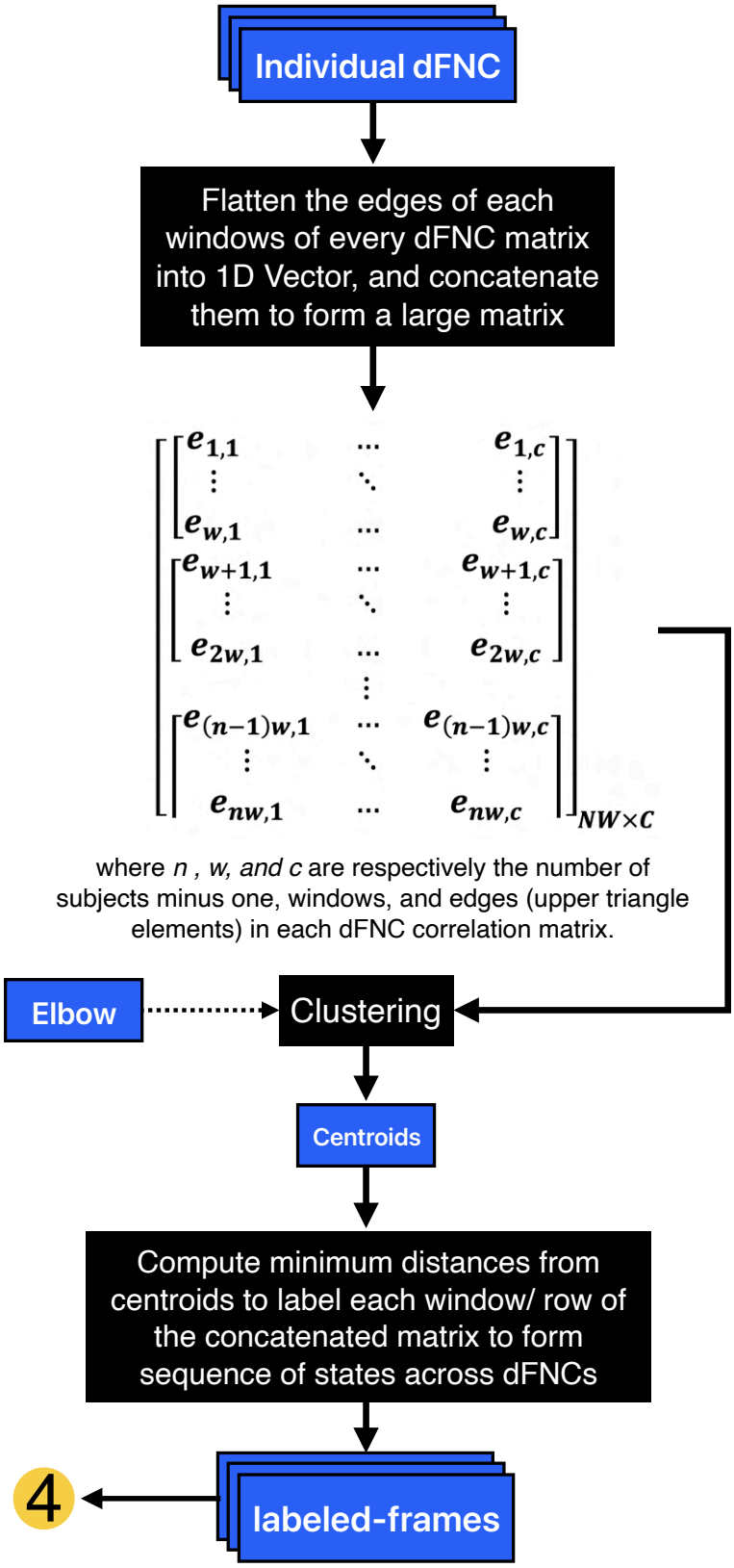
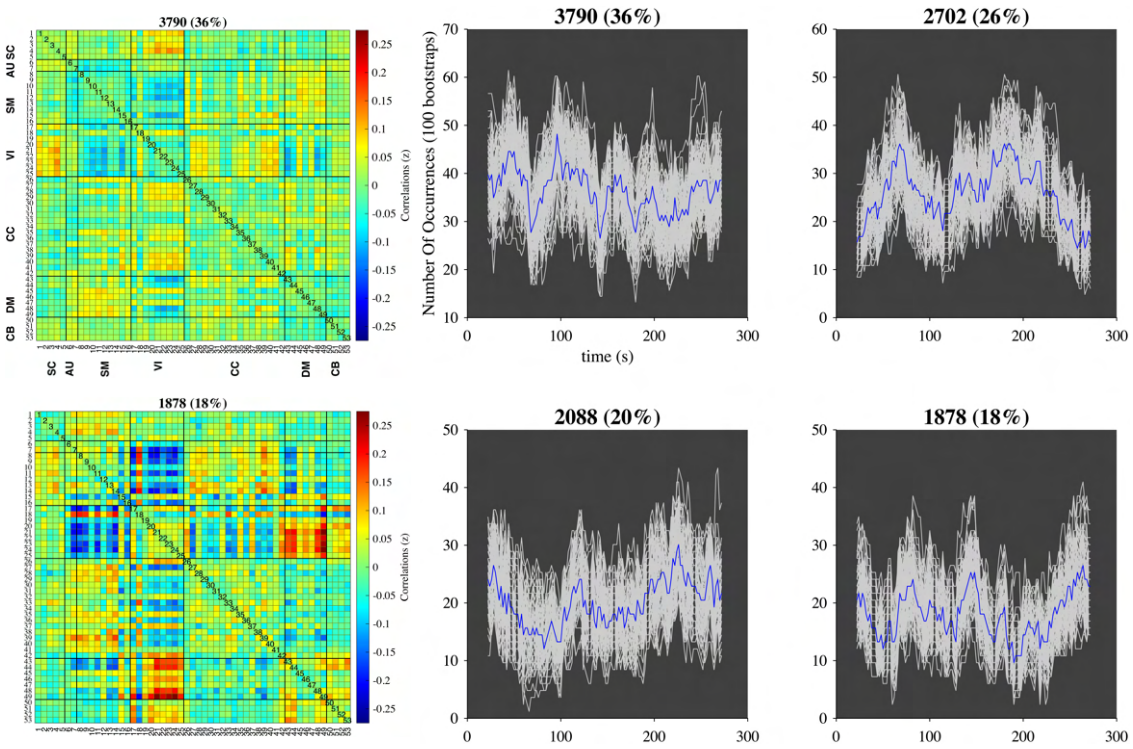


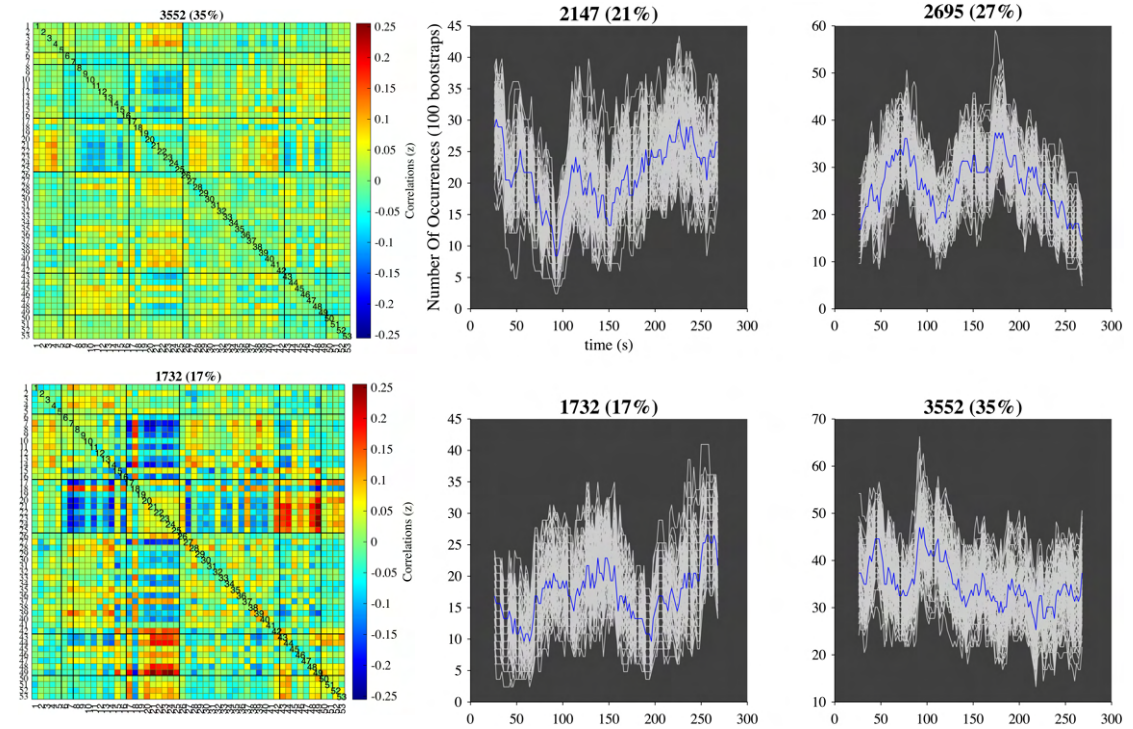
Figure 3: **Grouping windows and assigning labels (Step 3).** Each cluster centroid represented a distinct connectivity state, and the elbow criterion was used to determine the optimal number of states. In Step 4, frames were selected based on their assigned state labels.

Similarity of connectivity-state patterns across temporal scales

(a) 22–TRs



(b) 26–TRs



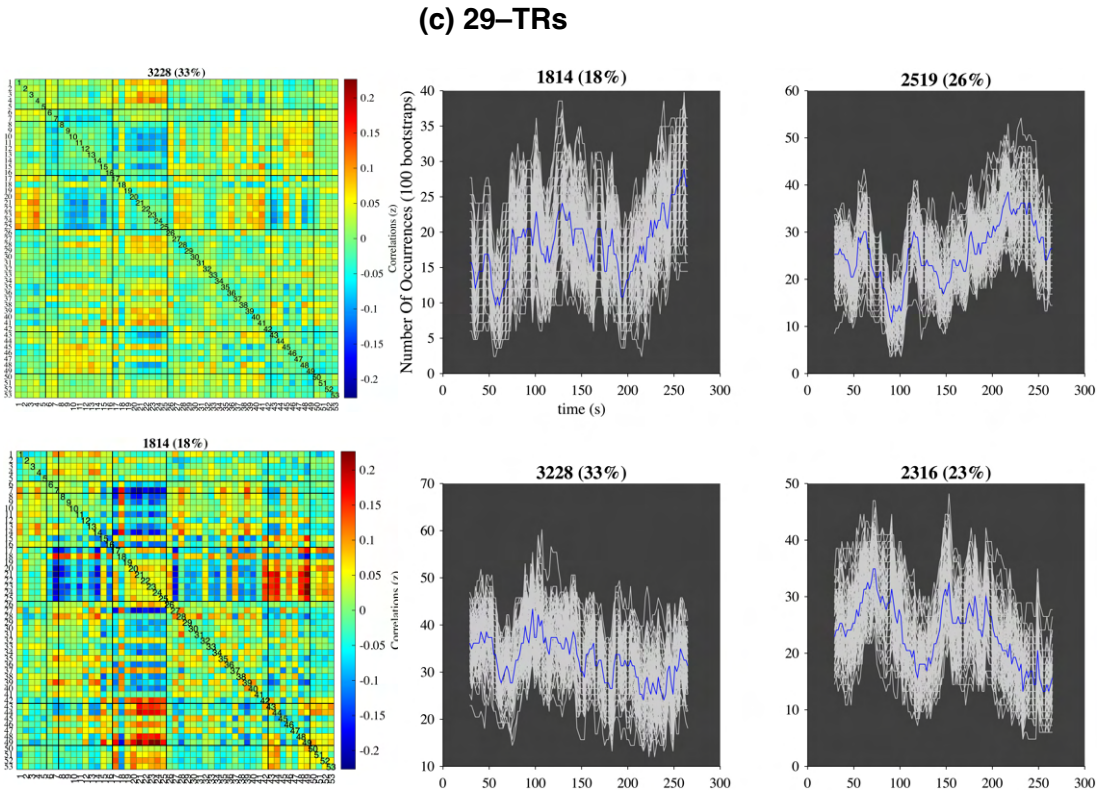


Figure 4: **Cluster centroids across temporal scales.** Connectivity patterns show similarities within and among the three most persistent states (States 1 in panel a, 4 in panel b, and 3 in panel c) and the three most transient states across the evaluated temporal scales.

Elbow criterion

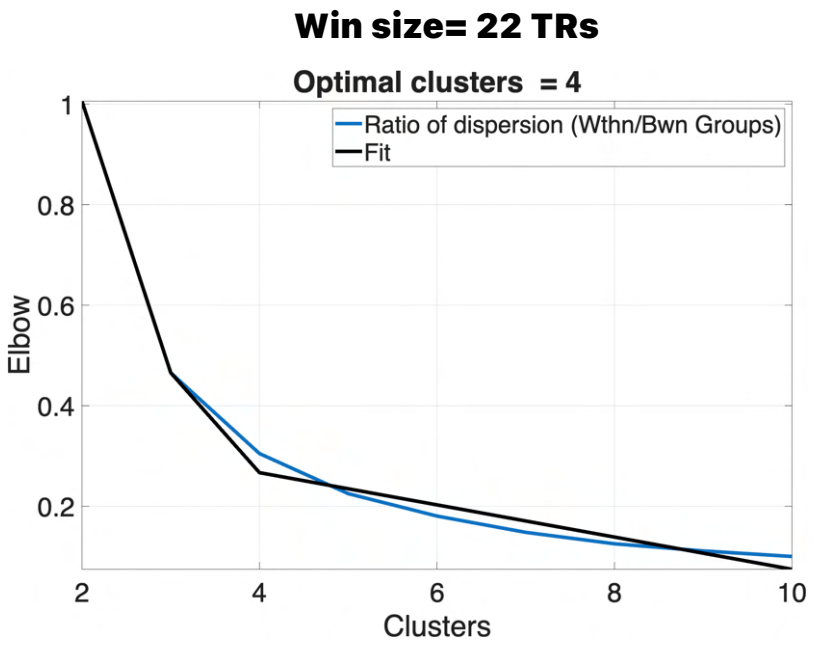


Figure 5: **Representative elbow plot.** The dispersion-ratio profiles supported (K=4) across all 11 temporal scales (20 to 30 TRs); only one representative plot is shown here. K-means clustering used *squared Euclidean* distance and 200 iterations.

References

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